GHOST PROTOCOL

Autonomous Cyber-Threat Intelligence, Entity Resolution & Graph Correlation Platform

System Design & Architecture Specification | Target Implementer: Antigravity | Status: Frozen ML Baseline

1. Executive Overview & Foundational Architecture

Ghost Protocol is an autonomous, production-grade Cyber Threat Intelligence (CTI) system built upon the frozen DarkWatch XAI machine learning engine. Moving far beyond isolated CVE pattern matching, Ghost Protocol operates as a continuous, memory-augmented intelligence platform capable of discovering, observing, resolving, and correlating complex threat entities across public, deep, and dark web sources.

Core Architectural Mandate

Ghost Protocol answers the operational question: *'What would a real-world autonomous CTI platform need to understand, remember, correlate, and investigate?'* It maintains absolute fidelity to our frozen DarkWatch ML models (DistilBERT, XGBoost, Temporal Autoencoder v2, HM-TIF Fusion), Whonix SOCKS5 fail-closed security boundary, and OASIS STIX 2.1 knowledge schemas.

2. System Subsystems & Agent Architecture

The Ghost Protocol platform comprises 10 tightly coupled, decoupled subsystems operating across asynchronous processing queues:

Subsystem Name	Core Tech / Boundary	Functional Responsibility
1. Discovery Engine	In-memory & DB Discovery Queue	Discovers target URLs via Daunt.link, Telegram previews, search indexers, and link hopping.
2. Whonix Collector	socks5h://10.152.152.10:9050	Executes fail-closed, JS-disabled HTTP requests & Playwright sessions over Tor isolation.
3. Preprocessor/Sanitizer	Trafilatura + MinHash LSH	Strips media, cleans HTML, masks PII/Crypto/Onion tags, and executes MinHash LSH deduplication.
4. Threat Pre-Filter	Regex Anchors + Density Filter	Filters generic chatter (<8 tokens), routing actionable CTI text payloads to ML models.
5. Frozen ML Pipeline	PyTorch & XGBoost Encoders	Infers text exploit type (DistilBERT), net transport (XGBoost), & temporal anomaly (Autoencoder v2).
6. HM-TIF Fusion Engine	Cross-Attention Fusion	Synthesizes multi-modal representations into a Dynamic Risk Score (0.00 - 1.00) with missing-data handling.

Subsystem Name	Core Tech / Boundary	Functional Responsibility
7. Entity Extractor (TIJERE)	Joint Regex/Transformer NLP	Extracts structured CTI entities: CVEs, IPs, Hashes, Wallets, Handles, Malware, & Access Claims.
8. Entity Resolver & Graph	Jaro-Winkler + Vector Cosine	Resolves recurring actors, disambiguates aliases, and maintains temporal graph memory.
9. Two-Tier XAI Engine	Cross-Attention + SHAP	Generates Tier 1 token highlights and Tier 2 post-hoc SHAP feature attribution bar charts.
10. Storage & API Layer	SQLite darkwatch.db + REST	Persists STIX 2.1 JSON bundles, provenance, and serves REST/gRPC endpoints for the SOC Dashboard.

3. Data Model & Storage Architecture

Ghost Protocol utilizes a relational SQLite storage engine (darkwatch.db) coupled with an in-memory NetworkX property graph to represent threat entities, STIX 2.1 domain objects, and temporal relationships.

Table Name	Column Schema	Purpose & Description
entities	entity_id (PK), canonical_name, entity_type, confidence_score, first_seen_utc, last_seen_utc, metadata_json	Tracks resolved threat entities (Actors, Usernames, Malware, CVEs, Orgs).
entity_aliases	alias_id (PK), entity_id (FK), alias_name, evidence_type, confidence, provenance_hash, created_utc	Stores alias mappings with supporting evidence and confidence scores.
entity_relationships	rel_id (PK), source_entity_id (FK), target_entity_id (FK), rel_type, confidence, provenance_json, created_utc	Property graph edges (e.g. USES, OPERATES, TARGETS, ASSOCIATED_WITH).
sources_registry	source_id (PK), canonical_url, source_type, validation_status, raw_html_sha256, clean_text_sha256, last_checked_utc	Tracks target platform health, discovery provenance, and MinHash signatures.
gold_records	record_id (PK), observation_id, clean_text_sha256, stix_bundle_json, mitre_tcodes_json, risk_score, xai_data_json	Stores fully enriched observation payloads, STIX 2.1 bundles, and XAI outputs.

4. Observation Lifecycle & Whonix Security Boundary

All real-world web communication is forcibly routed through the Whonix-Gateway SOCKS5 proxy (socks5h://10.152.152.10:9050) with remote DNS resolution. JavaScript is disabled by default for bulk crawling.

Lifecycle State	Trigger / Condition	Operational Action
UNCHECKED	URL queued from discovery engine.	Pre-flight socket check to 10.152.152.10:9050.
SERVICE_VALIDATED	HTTP GET over Whonix SOCKS5.	Classifies status: 200 OK, 503 Barrier, 401 Gated, 404 Dead.

Lifecycle State	Trigger / Condition	Operational Action
VERIFICATION_REQUIRED	Session-gated site (e.g. Dread).	Triggers authorized Playwright session to complete 2-step.
SESSION_ACTIVE	Session cookies in volatile vault.	Exports SHA-256 cookie hashes to log; injects cookies to httpx.
OBSERVING_CONTENT	Continuous JS-disabled fetch.	Executes Trafilatura clean & MinHash LSH deduplication.
PROCESSED	Observation sent downstream.	Dispatched to lab_bridge.py for ML risk scoring and CTI mapping.

5. Entity Taxonomy, Resolution & Alias Disambiguation

To prevent entity duplication (e.g. treating 10 observations of handle @DarkRider as 10 separate actors), Ghost Protocol implements an advanced **Entity Resolution Engine**:

Entity Type	Definition	Example Values
Company / Organization	Targeted enterprise, sector, or agency.	Acme Corp, Defense Ministry, BankX
Threat Actor / Alias	Individual handle, persona, or moniker.	@DarkRider, R3dDevil, Keymaster
Malware & Ransomware	Malicious software family or ransomware.	LockBit 3.0, Cobalt Strike, DarkStealer
Infrastructure & Crypto	IP address, domain, hash, or wallet.	192.168.1.1, 1A1zP1e... (BTC), SHA256
Access / Leak Claim	Stolen database claim or RCE listing.	50GB RDP access claim, SQLi database dump

Entity Resolution Algorithm & Confidence Matrix:

When a new entity string \(E_{new}\) is observed, the Entity Resolver computes a composite similarity score \(S_{comp}\) against existing canonical entities \(E_{canon}\):

Composite Resolution Formula:

$$S_{comp} = 0.40 \cdot JaroWinkler(E_{new}, E_{canon}) + 0.35 \cdot CosineSim(V_{new}, V_{canon}) + 0.25 \cdot GraphJaccard(N_{new}, N_{canon})$$

Score Range	Resolution Decision	System Action Taken
S_comp >= 0.85	AUTOMATIC MERGE	Merge E_new into E_canon; update last_seen_utc and increment co-occurrence count.
0.65 <= S_comp < 0.85	CANDIDATE ALIAS LINK	Create candidate alias edge in entity_aliases table with supporting evidence string.
S_comp < 0.65	NEW CANONICAL ENTITY	Instantiate new canonical entity in entities table with confidence = 0.90.

6. Temporal Graph Memory & Multi-Hop Correlation

Ghost Protocol maintains a temporal property graph allowing multi-hop intelligence queries across historical observations.

Edge Relationship Type	Semantic Meaning	Concrete Instance Example
(Actor) --[OPERATES]--> (Infrastructure)	Threat actor controls IP / domain / wallet.	Actor @DarkRider operates wallet 1A1z...
(Actor) --[USES]--> (Malware)	Threat actor deploys specific ransomware.	Actor @DarkRider uses LockBit 3.0
(Actor) --[USES_ALIAS]--> (Alias)	Actor linked to alternate handle.	Actor @DarkRider uses alias R3dDevil
(Malware) --[TARGETS]--> (Organization)	Ransomware campaign targets company.	LockBit 3.0 targets Acme Corp
(Observation) --[EVIDENCE_FOR]--> (Claim)	Post contains proof of data breach.	Obs_9f8b evidence for 50GB Leak Claim

7. Frozen AI Pipeline Integration & Two-Tier XAI

The machine learning brain remains 100% frozen. Incoming sanitized text, 80 network flow features, and 14 temporal features are processed without model retraining:

ML Component	Functional Role in Pipeline	Performance / Config
HackerSignal DistilBERT	Text Exploit Category (8 classes: injection, xss, rce, etc.)	Accuracy: 91.30%, Macro-F1: 89.07%
Multiclass XGBoost	Encrypted Network Transport Mode (Clearnet, Tor, VPN)	79 Flow Features
Temporal Autoencoder v2	Behavioral Anomaly MSE Reconstruction Loss	14 Features -> 8-dim Latent Space z
HM-TIF Fusion Engine	Dynamic Cross-Attention Risk Score (0.00 - 1.00)	Handles missing modalities (e.g. 503 barrier)
Two-Tier XAI Generator	Tier 1 Token Highlights + Tier 2 Post-Hoc SHAP Bar Charts	Triggered automatically when Risk Score >= 0.70

8. Analyst Workstation & On-Demand OSINT

The SOC Workstation interface exposes interactive **Explained Alert Cards**, STIX 2.1 bundle exports, CTI entity search, and **On-Demand Sherlock Handle Lookups**:

Workstation Feature	Analyst Action Triggered	System Execution Outcome
CTI Indicator Search	Query CVE, malware, hash, or org string.	Returns matching STIX bundles, source URLs, timestamps, & XAI.
On-Demand Sherlock	Click 'Lookup Handle via Sherlock' on user.	Executes transient OSINT query across public sites; NOT stored in DB.

Workstation Feature	Analyst Action Triggered	System Execution Outcome
STIX 2.1 Bundle Export	Click 'Export STIX 2.1 JSON' button.	Downloads OASIS-compliant JSON bundle for SIEM/SOAR ingestion.
Human Verification	Review candidate alias links (S_comp < 0.85).	Analyst confirms/rejects proposed alias merges.

9. Implementation Blueprint for Antigravity

Antigravity will execute implementation by creating and modifying the following repository modules in exact order:

Module Path	Type	Implementation Responsibilities
darkwatch/discovery/source_registry.py	NEW	SQLite manager for sources_registry, entities, entity_aliases, and entity_relationships.
darkwatch/discovery/search_discoverer.py	NEW	Parses Daunt.link, Ahmia dark web indexer, and deepdarkCTI GitHub feeds.
darkwatch/observer/telegram_public_observer.py	NEW	HTML parser targeting t.me/s/SliceForLifeee public web preview without login.
darkwatch/observer/dread_session_observer.py	NEW	Playwright SOCKS5 session manager for Dread 2-step verification and cookie vault.
darkwatch/resolution/entity_resolver.py	NEW	Jaro-Winkler + Cosine + Graph Jaccard resolution engine for alias disambiguation.
darkwatch/investigation/sherlock_wrapper.py	NEW	Transient on-demand handle lookup wrapper.
darkwatch/pipeline/lab_bridge.py	MODIFY	Chains Stage 1 observations into frozen ML pipeline, STIX, MITRE, XAI, and darkwatch.db.
darkwatch/dashboard/app.py	NEW	Flask/Streamlit SOC Dashboard rendering Explained Alert Cards and graph view.

System Design Sign-Off: *This document represents the complete, frozen specification for Ghost Protocol v1.0.0-PROD. Ready for Antigravity code generation.*